

Simulation Parameters, Expectancy Modeling, and Design-Specific Effects in N-of-1 Trial Simulations

A Comprehensive Analysis of pmsimstats2025 Architecture

Statistical Simulation Series

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1 Introduction

The pmsimstats2025 project simulates N-of-1 clinical trials with emphasis on detecting biomarker-moderated treatment effects in the presence of carryover effects. The simulation architecture consists of three integrated components: (1) parameter definitions that specify study conditions, (2) a three-factor response model (Biological, Expectancy, Time-variant) aligned with Hendrickson et al. (2020), and (3) design-specific measurement schedules that differentially impact how expectancy confounds effect detection.

This white paper systematically addresses three interconnected questions:

1. **What parameter settings impact simulation results?** A comprehensive inventory of tunable, fixed, and derived parameters
2. **How does expectancy enter the simulation, and what is its relationship to Hendrickson methodology?** The theoretical framework and implementation
3. **Does expectancy impact trial designs differently?** A comparative analysis of OL+BDC versus Hybrid designs

The central finding is that **expectancy (ER) component has differential confounding potential across designs**, with OL+BDC being significantly more vulnerable to ER-BR coupling effects than the Hybrid design, due to timing of expectancy drop relative to treatment randomization.

2 Part I: Parameter Settings and Their Impact on Simulation Results

2.1 1.1 Parameter Taxonomy

Simulation parameters fall into five categories with distinct roles in determining results:

2.1.1 1.1.1 Simulation Control Parameters (Fixed, Non-Tunable)

These parameters determine computational scope and reproducibility:

Parameter	Value	Purpose
<code>n_participants</code>	70	Sample size per Monte Carlo iteration
<code>n_iterations</code>	100 (clustered), 1000 (evenly-spaced)	Monte Carlo replicates per condition
Random seed	<code>set.seed(iter * 1000 + i)</code>	Deterministic but varied seeds

Impact on Results:

- Increasing `n_iterations` reduces standard error of power estimates ($\sim$ proportional to $1/\sqrt{n}$)
- Larger `n_participants` reduces variance within each iteration but increases computation

2.1.2 1.1.2 Data Generation: Fixed Effect Sizes (Baseline Parameters)

These parameters define the **scale** of responses and are held constant across all simulations:

Parameter	Value	Interpretation
BR_rate	0.5	Biological response improvement (points/week on treatment)
ER_rate	0.2	Expectancy response improvement (points/week with expectancy)
TR_rate	0.1	Time-variant natural improvement (points/week elapsed)
baseline_mean	10.0	Mean baseline symptom severity
biomarker_mean	5.0	Mean biomarker level
between_subject_sd	2.0	SD of random intercepts
within_subject_sd	1.8	SD of measurement noise
biomarker_sd	2.0	SD of biomarker distribution

Impact on Results:

- These are **NOT varied** across simulation conditions
- Changing any would **proportionally scale all power estimates** but not change relative patterns
- Current values derived from Hendrickson et al. (2020) preliminary data

2.1.3 1.1.3 Correlation Structure Parameters (Fixed, Optimized)

These parameters specify the **covariance matrix** structure and are held constant to maintain positive definiteness:

Autocorrelations (within-component temporal correlation):

- `c.br` = 0.75 (biological response, reduced from Hendrickson's 0.8)
- `c.er` = 0.75 (expectancy response, reduced from 0.8)
- `c.tr` = 0.75 (time-variant response, reduced from 0.8)

Cross-Correlations (between-component same or different timepoint):

- `c.cf1t` = 0.12 (same-timepoint BR-ER-TR correlation, reduced from 0.2)
- `c.cfct` = 0.05 (different-timepoint cross-correlation, reduced from 0.1)
- `c.bm_baseline` = 0.25 (biomarker-baseline correlation, reduced from 0.3)
- `c.baseline_resp` = 0.3 (baseline-response correlation, reduced from 0.4)

Biomarker Interaction Correlation (dynamic):

- `c.bm` = 0.3 when participant on treatment
- `c.bm` = 0 when participant off treatment

Why Fixed:

- Original Hendrickson values (`c.br=0.8`, `c.cf1t=0.2`, `c.cfct=0.1`) created non-positive definite matrices with 8-point clustered schedules
- Balanced reduction maintains correlation hierarchy while ensuring PD for all parameter combinations
- Tested via `validate_parameter_grid()` pre-simulation validation

Impact on Results:

- Correlation parameters affect **signal-to-noise ratio** and **temporal patterns**
- Changes would require re-validating all 3 sigma structures before simulation
- Minimal exploratory value (parameters determined by theory, not data-driven)

2.1.4 1.1.4 Parameter Grid: Variable Conditions (Tuned Across Simulations)

These parameters define the **combinations tested** in each simulation:

```
param_grid <- bind_rows(  
  expand_grid(  
    design = "ol_bdc",  
    biomarker_moderation = c(0.25, 0.35, 0.45, 0.55, 0.65),  
    biomarker_correlation = c(0.3),  
    carryover = c(0, 0.5, 1)  
  ),  
  expand_grid(  
    design = "ol_bdc",  
    biomarker_moderation = c(0),  
    biomarker_correlation = c(0),  
  )  
)
```

```

    carryover = c(0, 0.5, 1)
  )
)

```

2.1.4.1 OL+BDC Design Parameter Grid

2.1.4.2 Parameter Definitions and Impact biomarker_moderation (HIGHEST IMPACT)

- **Range tested:** 0 to 0.65
- **Formula:** $\text{effective_BR_rate} = \text{BR_rate} * (1 + \text{biomarker_moderation} * \text{biomarker_centered})$
- **Interpretation:** At each SD increase in biomarker, BR rate increases by this proportion
- **Impact on power: Linear relationship** — higher moderation → higher power to detect interaction

biomarker_correlation / c.bm (MODERATE IMPACT)

- **Values tested:** 0 (null), 0.3 (effect)
- **Role:** Controls covariance structure between biomarker and biological response
- **Mechanism:** When on treatment: $\text{Cor}(\text{biomarker}, \text{BR}) = \text{c.bm}$; When off: $\text{Cor}(\text{biomarker}, \text{BR}) = 0$
- **Impact:** Creates **implicit interaction** through correlation structure (Hendrickson approach)

carryover / t1half (MODERATE IMPACT)

- **Standard implementation:** $\text{carryover} \in \{0, 0.5, 1.0\}$ (fixed proportion retention)
- **Alternative implementation:** $\text{t1half} \in \{0, 1.0, 2.0, 3.0\}$ (exponential decay half-life)
- **Impact on power:** $\text{carryover} = 0$ (no effect), 0.5 (10-20% reduction), 1.0 (20-40% reduction)
- **Confounding mechanism:** Carryover adds unexplained variance to off-period measurements

design (HIGHEST IMPACT TIER 2)

- **Options:** “hybrid”, “ol_bdc” (clustered designs); “ol”, “crossover”, “parallel” (evenly-spaced)
- **Measurement schedules:**
 - Hybrid: $\text{c}(4, 8, 9, 10, 11, 12, 16, 20)$ — 8 points with clusters
 - OL+BDC: $\text{c}(4, 8, 12, 16, 17, 18, 19, 20)$ — 8 points with late cluster
 - OL, Crossover, Parallel: $\text{c}(2, 8, 14, 20)$ — 4 points (6-week intervals)
- **Impact:** Fundamental effect on matrix conditioning, expectancy confounding, and statistical power

2.2 1.2 Summary: Which Parameters Have Largest Impact?

2.2.1 Rank Ordering by Effect on Power Estimates

Rank	Parameter	Impact Magnitude	Recommendation
1	design (structure)	Varies power by 0-30%	Core comparison variable
2	biomarker_moderation	Linear power increase	Primary variable of interest
3	carryover	Reduces power by 10-40%	Key confound mechanism
4	biomarker_correlation (c.bm)	Enables interaction	Structural requirement
5	n_iterations	Reduces SE, not mean	Precision parameter
6	n_participants	Affects per-iteration variance	Secondary precision

Practical guidance:

- To explore new scenarios: **vary design, biomarker_moderation, carryover**
- To validate findings: **increase n_iterations and n_participants**
- **Do not change:** correlation parameters, effect sizes, or measurement schedules (requires re-validation)

3 Part II: Expectancy Modeling and Relationship to Hendrickson Methodology

3.1 2.1 The Three-Factor Response Model: BR, ER, TR

Building on Hendrickson et al. (2020), the pmsimstats2025 simulation decomposes observed response into three independent components:

$$\text{Response}_t = \text{Baseline} + \text{BR}_t + \text{ER}_t + \text{TR}_t + \epsilon_t$$

where $\epsilon_t \sim N(0, \sigma^2)$ is measurement noise.

3.1.1 2.1.1 Biological Response (BR) — The Drug Effect

Definition: Improvement in symptoms due to drug mechanism of action.

Formula:

$$BR_t = w_t \cdot r_{\text{eff}} + \text{Carryover}_t$$

where: - w_t = cumulative weeks on active treatment - $r_{\text{eff}} = r_0(1 + \beta_{\text{mod}} \cdot BM_{\text{centered}})$ = **biomarker-moderated** rate - r_0 = baseline rate (BR_rate = 0.5 points/week) - β_{mod} = biomarker moderation (0 to 0.65 in simulations) - Carryover_t = residual effect during off-treatment periods

Code Implementation:

```
BR_mean = {
  effective_BR_rate <- BR_rate * (1 + bm_mod * bm_centered)
  br_accumulated <- lag(weeks_on_drug, default = 0) * effective_BR_rate
  first_off <- treatment == 0 & lag(treatment, default = 1) == 1
  ifelse(treatment == 1,
    weeks_on_drug * effective_BR_rate,
    ifelse(first_off, br_accumulated * params$carryover, 0)
  )
}
```

Key Properties:

1. **Linear accumulation during treatment:** BR grows proportionally with time on drug
2. **Biomarker-moderated:** Higher biomarkers $\rightarrow$ stronger drug response
3. **Subject to carryover:** Persists after discontinuation through exponential or fixed-proportion decay
4. **Temporally correlated:** Autocorrelation $c.br = 0.75$ (AR(1) structure)

3.1.2 2.1.2 Expectancy Response (ER) — The Contextual Effect

Definition: Improvement due to psychological/contextual factors, varying with **knowledge of treatment** status.

Formula:

$$ER_t = w_{\text{expect},t} \cdot \gamma_{\text{ER}} + \text{random}_{\text{ER}}$$

where: - $w_{\text{expect},t}$ = cumulative weeks with expectancy (blinding status) - γ_{ER} = expectancy response rate (ER_rate = 0.2 points/week)

Expectancy Assignment:

- **OL+BDC:** expectancy = if_else(week <= 16, 1.0, 0.5)
 - Weeks 0-16: Open-label $\rightarrow$ expectancy = 1.0
 - Weeks 17-20: Blinded discontinuation $\rightarrow$ expectancy = 0.5
- **Hybrid:** expectancy = if_else(week %in% c(4, 8), 1.0, 0.5)
 - Weeks 4, 8: Early open-label measurements $\rightarrow$ expectancy = 1.0
 - All other weeks: Blinded $\rightarrow$ expectancy = 0.5

Code Implementation:

```
weeks_with_expectancy = cumsum(expectancy),  
ER_mean = weeks_with_expectancy * ER_rate,  
ER = ER_mean + er_random,
```

Key Properties:

1. **Design-driven:** Determined by blinding status, not treatment exposure
2. **NOT biomarker-moderated:** Placebo effects are typically non-specific
3. **Cumulative:** Expectancy weeks accumulate continuously
4. **No carryover:** ER resets when expectancy knowledge changes (unlike BR carryover)

3.1.3 2.1.3 Time-variant Response (TR) — Natural Trajectory

Definition: Symptom improvement due to disease natural history, independent of any intervention.

Formula:

$$TR_t = t_{\text{trial}} \cdot \rho_{\text{TR}} + \text{random}_{\text{TR}}$$

where: - t_{trial} = weeks elapsed since trial start - ρ_{TR} = time-variant rate (TR_rate = 0.1 points/week)

Final Assembly:

```
response = baseline + BR + ER + TR
```

3.2 2.2 Why Expectancy is Modeled Separately: Hendrickson's Methodological Insight

3.2.1 2.2.1 The N-of-1 Confounding Problem

In standard **parallel-group RCTs**: Between-group structure naturally separates BR (drug) from ER (placebo).

In **N-of-1 trials**: Single participant experiences both treatment and control. Without explicit modeling, BR and ER become confounded within-person.

Hendrickson et al. (2020) solved this by explicitly decomposing the response into three factors, enabling researchers to:

1. Estimate how much improvement is drug-driven (BR)
2. Estimate how much is contextual (ER)
3. Detect biomarker-moderated **BR** specifically (not ER, which is non-specific)

3.2.2 2.2.2 Design Advantage: Hybrid Strategy Exploits Factor Separation

The **Hybrid N-of-1 design** strategically uses the 3-factor decomposition:

Phase 1: Open-Label Run-In (Weeks 0-8)

- Everyone on active treatment, unblinded
- BR and ER both accumulate strongly

- **Purpose:** Establish tolerance, get baseline on-drug response

Phase 2: Randomized Discontinuation (Weeks 9-12)

- Participants randomized to continue or discontinue (blinded)
- **Paths A,B (continue):** BR stays high, ER drops (blinded, lose open-label advantage)
- **Paths C,D (discontinue):** BR drops via carryover, ER drops (blinded)
- **Key insight:** Both BR and ER drop uniformly across paths when blinded

Phase 3: Blinded Crossover (Weeks 13-20)

- All participants blinded, everyone crosses over
- **ER is now constant** across treatment and control
- **BR varies** by treatment status
- **Result:** Pure BR×biomarker interaction can be detected with ER as a constant background

3.2.3 2.2.3 Enhancement: Making Expectancy Explicit

Current pmsimstats2025 Enhancement:

```
expectancy = if_else(week <= 16, 1.0, 0.5) # OL+BDC
expectancy = if_else(week %in% c(4, 8), 1.0, 0.5) # Hybrid
```

Why this matters:

1. **More realistic:** Placebo effects should diminish when participants learn their treatment status
2. **More interpretable:** Directly models the blinding-induced difference
3. **Better design evaluation:** Can assess how much unblinding in early phases affects later power

3.3 2.3 The Covariance Structure: How the Three Components Correlate

The three response components are **not independent**—they’re drawn from a **multivariate normal distribution** with explicit correlations.

3.3.1 Autocorrelations (Within-Component)

Each component exhibits **first-order autoregressive (AR(1)) structure** with decay parameter 0.75:

$$\text{Cor}(\text{Component}_t, \text{Component}_{t+\Delta}) = c_{\text{component}}^{\Delta}$$

For BR component: - Cor(BR at week 4, BR at week 8) = $0.75^1 = 0.75$ - Cor(BR at week 4, BR at week 12) = $0.75^2 = 0.5625$ - Cor(BR at week 4, BR at week 20) = $0.75^4 = 0.316$

3.3.2 Cross-Correlations (Between-Component)

Different components correlate modestly:

- `c.cf1t = 0.12` (same-timepoint cross-correlation)

- `c.cfct` = 0.05 (different-timepoint cross-correlation)

3.3.3 Biomarker Interaction Correlation (Dynamic)

Critical for modeling biomarker×treatment interaction:

$$\text{Cor}(\text{Biomarker}, \text{BR}) = \begin{cases} c.bm = 0.3 & \text{if Treatment} = 1 \\ 0 & \text{if Treatment} = 0 \end{cases}$$

Mechanism: When on treatment, biomarker level predicts BR response. When off treatment, both revert to baseline.

4 Part III: Differential Impact of Expectancy on OL+BDC vs. Hybrid Designs

4.1 3.1 Measurement Schedules and Expectancy Allocation

4.1.1 OL+BDC Design

Measurement weeks: `c(4, 8, 12, 16, 17, 18, 19, 20)` — 8 points

Expectancy rule: `expectancy = if_else(week <= 16, 1.0, 0.5)`

Week	Measure	Expectancy	Phase	ER_accumulated
4		1.0	Open-label	0.2
8		1.0	Open-label	0.4
12		1.0	Open-label	0.6
16		1.0	Last open-label	0.8
17		0.5	Blinded (drop)	0.9
18		0.5	Blinded	1.0
19		0.5	Blinded	1.1
20		0.5	Blinded	1.2

4.1.2 Hybrid N-of-1 Design

Measurement weeks: `c(4, 8, 9, 10, 11, 12, 16, 20)` — 8 points

Expectancy rule: `expectancy = if_else(week %in% c(4, 8), 1.0, 0.5)`

Week	Measure	Expectancy	Phase	ER_accumulated
4		1.0	Open-label	0.2
8		1.0	Open-label	0.4
9		0.5	Blinded (drop)	0.5
10		0.5	Blinded discontinuation	0.6
11		0.5	Blinded discontinuation	0.7
12		0.5	Blinded discontinuation	0.8

Week	Measure	Expectancy	Phase	ER_accumulated
16		0.5	Blinded crossover	1.0
20		0.5	Blinded crossover	1.2

4.2 3.2 Hypothesis Evaluation: “Expectancy Impacts Designs Differently”

4.2.1 Hypothesis Status: CONFIRMED WITH NUANCE

Your conjecture is correct: **OL+BDC** is significantly more vulnerable to expectancy confounding.

4.3 3.2.1 Dimension 1: Duration of High-Expectancy Phase

Design	High ER Phase	Measurement Weeks	Duration
OL+BDC	Weeks 0-16	4 weeks	16 weeks
Hybrid	Weeks 0-8	2 weeks	8 weeks

Finding: OL+BDC maintains high expectancy for **twice as long**, creating a prolonged phase where BR and ER are strongly coupled.

4.4 3.2.2 Dimension 2: Expectancy Drop Timing Relative to Randomization

OL+BDC Confounding Pattern: “Coupled Decline”

- Weeks 16→17: Both randomization AND expectancy drop
- Paths 3,4 (discontinue): BR drops (carryover), ER drops (unblinded)
- **Cannot cleanly separate** BR loss from ER loss

Hybrid Confounding Pattern: “Independent Variation”

- Week 8→9: Expectancy drops BEFORE randomization (week 10)
- When paths randomize, they have the **same ER level** (0.5)
- Paths diverge in BR while ER is already equalized
- **Clean separation** of BR effect from ER noise

4.5 3.2.3 Dimension 3: Measurement Density in Blinded Phase

Design	Blinded Measurements	Structure
OL+BDC	Weeks 17, 18, 19, 20 (4 consecutive weeks)	Between-subjects
Hybrid	Weeks 9, 10, 11, 12, 16, 20 (6 measurements)	Within-subjects (crossover)

Implication:

- OL+BDC: Dense but no crossover → harder to isolate BR
- Hybrid: Within-subject design → each subject serves as own control, ER held constant

4.6 3.3 Power Implications

4.6.1 OL+BDC Power Trajectory

- **Weeks 4-16 (Open-label):** Power inflated by ER coupling
- **Weeks 17-20 (Blinded):** Power moderate, confounded estimates
- **Overall:** Power peaks mid-trial, then declines

4.6.2 Hybrid Power Trajectory

- **Weeks 4-8 (Open-label):** Power inflated by ER coupling
- **Weeks 9-12 (Blinded with ER controlled):** Power stable and unbiased
- **Weeks 16, 20 (Crossover):** Power high from within-subject design
- **Overall:** Power high and stable in critical measurement windows

4.7 3.4 Summary: Expectancy Vulnerability Assessment

4.7.1 Vulnerability Ranking

Design	Expectancy Vulnerability	Mechanism
OL+BDC	High	Long high-ER phase, coupled BR-ER drop
Hybrid	Low	Short high-ER phase, ER drops before randomization
Crossover	Very Low	All blinded, ER constant throughout

Design	Expectancy Vulnerability	Mechanism
Parallel	High	All open-label, ER high in both groups

5 Conclusion and Recommendations

5.1 Summary of Findings

5.1.1 1. Parameter Impact Ranking

The most influential parameters are:

1. **Design structure** (fundamental effect on power, confounding, measurement density)
2. **Biomarker_moderation** (primary effect size, linear power relationship)
3. **Carryover** (10-40% power reduction, confounding mechanism)
4. **Biomarker_correlation/c.bm** (enables two-level interaction structure)

5.1.2 2. Expectancy in Simulation

Following Hendrickson et al. (2020), the pmsimstats2025 simulation explicitly models expectancy (ER) as a separate response component, varying with **blinding status**. This enables:

- Separation of psychological (ER) from pharmacological (BR) effects
- Design-specific control of expectancy confounding
- Proper power estimation for biomarker×BR interactions

5.1.3 3. Design-Specific Expectancy Vulnerability

OL+BDC is significantly more vulnerable to expectancy confounding due to:

- Longer high-ER phase (16 weeks vs. 8 weeks for Hybrid)
- Simultaneous BR and ER drop at randomization (confounded comparison)
- 4-week blinded phase may be insufficient to isolate BR from ER background

Hybrid design is more robust because:

- ER drops (week 8→9) before randomization (week 10)
- Paths diverge in BR while ER is already equalized
- Within-subject crossover (weeks 16, 20) provides clean BR comparison

Recommendation: For unbiased biomarker×BR interaction detection, **Hybrid design is superior**. OL+BDC is better for exploratory sampling where signal amplification (BR + ER) aids responder identification.

6 References

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